Digital Computer Organization

Examination Solutions (E1PA107T)

Section A: Combinational & Sequential Circuits

1. What is a Decoder? Explain the working of a 2-to-4 line Decoder.

A decoder is a combinational circuit that converts binary information from 'n' input lines to a maximum of 2^n unique output lines. For a 2-to-4 decoder, there are two inputs (A, B) and four outputs (D0, D1, D2, D3).

Truth Table:

A	B	D0	D1	D2	D3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

2. Half Adder vs. Full Adder

- **Half Adder:** Adds two single-bit binary numbers. It has 2 inputs and 2 outputs (Sum, Carry).
- **Full Adder:** Adds three single-bit binary numbers (two bits and one carry-in). It has 3 inputs and 2 outputs (Sum, Carry-out).

4. SR Flip-Flop characteristic equation.

The SR (Set-Reset) Flip-Flop is a basic sequential element. Its characteristic equation is:

Q(n+1) = S + R'Q(n) with the constraint that **SR = 0** (to avoid the forbidden state).

Section B: Register Organization & Micro-operations

6. Types of Micro-operations

- **Arithmetic:** Perform numeric calculations (e.g., `R1 ← R1 + R2`).
- **Logic:** Perform bitwise operations (e.g., `R1 ← R1 AND R2`).
- **Shift:** Used for serial data transfer (e.g., `shl R1`).
- **Control:** Manage the flow of instructions (e.g., `Conditional Branching`).

8. Shift Register Modes

- **SISO:** Serial-In Serial-Out
- **SIPO:** Serial-In Parallel-Out
- **PISO:** Parallel-In Serial-Out
- **PIPO:** Parallel-In Parallel-Out

Section C: CPU Registers & Boolean Minimization

9. Types of CPU Registers

Common registers include the **Program Counter (PC)** which holds the next instruction address, the **Accumulator (AC)** for general processing, the **Instruction Register (IR)** for current instruction storage, and **Address/Data Registers** for memory interfacing.